

POLARITON BLADE DEVICE

MATERIALS AND ENGINEERING SPECIFICATION

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FIELD	VALUE
Document Title	Polariton Blade Device — Materials and Engineering Specification
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1. SCOPE

This document specifies materials, dimensional requirements, operating parameters, and known engineering constraints for a polariton condensate blade device. All specifications are derived from published experimental data. Parameters marked [REQUIRES PROTOTYPE MEASUREMENT] have not been empirically established at device scale and must be determined during prototype development.

This is an engineering specification. Theoretical background is not included here. See companion paper for physics derivation.

2. COMPONENT ARCHITECTURE

The device consists of five components:

#	COMPONENT	FUNCTION	FABRICATION STATUS
1	Crystal Substrate	Phonon-photon coupling — generates polariton condensate	Lab scale demonstrated
2	Emitter Geometry	Shapes condensate output mode into blade waveguide	Requires nanofab tooling
3	Initiation Coil	EM field drives phonon cascade in crystal	Prototype feasible
4	Blade Waveguide	SWCNT lattice — confines and propagates condensate	Centimeter scale demonstrated
5	Hilt Assembly	Housing, power delivery, activation interface	Standard engineering

3. COMPONENT 1 — CRYSTAL SUBSTRATE

3.1 Material

PARAMETER	SPECIFICATION
Primary material	Cesium lead bromide perovskite — CsPbBr3
Crystal form	Single crystal — microwire or microplatelet geometry
Synthesis method	Template-assisted fabrication (Kedziorowa et al. 2024 method)
Crystal quality requirement	Minimal nonradiative recombination sites — high photoluminescence quantum yield req
Polarity	All-inorganic — no organic cation

3.2 Optical Parameters (Confirmed — CsPbBr3)

PARAMETER	VALUE	SOURCE
Emission wavelength	~520 nm (green)	Su et al. 2018
Exciton binding energy	~40 meV	Su et al. 2018
Polariton condensation threshold (CW)	~0.53 W/cm²	Wu et al. 2025
Polariton propagation distance (RT)	>60 micrometers	Su et al. 2018
Condensate linewidth	~0.5 meV	Wu et al. 2025
Operating temperature	Room temperature (confirmed)	Multiple 2024-2025 sources

3.3 Dimensional Requirements

PARAMETER	TARGET	STATUS
Crystal length	[REQUIRES PROTOTYPE MEASUREMENTS]	Depends on coil geometry
Crystal cross-section	[REQUIRES PROTOTYPE MEASUREMENTS]	Depends on emitter output mode
Emitter face diameter	Match to blade waveguide inner diameter	Requires nanofab
Surface roughness — emitter face	Sub-wavelength smoothness required	FIB milling target

NOTE: Blade color is determined by crystal lattice geometry and composition. CsPbBr3 emits green (~520 nm). Other halide perovskite compositions shift emission wavelength. Color is not separately adjustable — it is set by crystal selection.

4. COMPONENT 2 — EMITTER GEOMETRY

4.1 Specification

PARAMETER	SPECIFICATION
Material	Same crystal as Component 1 — continuous structure, no interface
Function	Shape polariton condensate exit mode to match blade waveguide inner diameter
Geometry	Cylindrical output mode — diameter matched to waveguide
Fabrication method	Focused ion beam (FIB) milling
Critical constraint	Zero material discontinuity between crystal body and emitter face — any interface = dec

NOTE: The emitter is not a separate component. It is the terminal end of the crystal, shaped to specification. Separate emitter materials are NOT acceptable.

5. COMPONENT 3 — INITIATION COIL

5.1 Specification

PARAMETER	SPECIFICATION
Geometry	Solenoid — wound around crystal body within hilt
Conductor material	High conductivity — copper or silver preferred
Target frequency	Phonon resonance frequency of specific crystal batch [REQUIRES MEASUREMENT]
Drive signal type	Continuous wave — not pulsed
Drive power	Sufficient to maintain condensate above threshold (~0.53 W/cm² baseline) [REQUIRES MEASUREMENT]
Tuning method	Spectroscopic measurement of crystal phonon resonance post-synthesis — coil wound around crystal
Coil-crystal pairing	1:1 — each coil is tuned to one specific crystal. Cross-compatibility is NOT expected.

NOTE: Coil frequency must be measured and matched to the specific crystal. Generic coil specifications cannot be provided — they are crystal-dependent. This is a physical constraint, not a design choice.

6. COMPONENT 4 — BLADE WAVEGUIDE

6.1 Material

PARAMETER	SPECIFICATION
Primary material	Single-walled carbon nanotubes (SWCNT) — semiconducting chirality, polymer-sorted
Preferred SWCNT type	(6,5) chirality — demonstrated polariton coupling, near-IR operation (Graf et al. 2016)
Structure	Cylindrical composite lattice — SWCNT film on structural substrate
Optical requirement	Translucent — condensate emission must be visible through lattice wall
Phonon conductance	Ballistic phonon transport along tube axis — inherent to SWCNT geometry

6.2 Confirmed Performance Parameters (SWCNT Polariton Systems)

PARAMETER	VALUE	SOURCE
Rabi splitting	>110 meV	Graf et al. 2016
Coupling strength ratio	Up to 13.3%	Multiple sources
Operating temperature	Room temperature (confirmed)	Graf et al. 2017
Electrical tunability	Confirmed via gate voltage	Graf et al. 2017
Polariton propagation velocity	Up to 3×10 ⁸ m/s	ACS Photonics 2021

6.3 Dimensional Requirements

PARAMETER	TARGET	STATUS
Inner diameter	Match to crystal emitter output mode diameter	Crystal dependent
Wall thickness	Minimum for structural integrity — measured post-synthesis	Requires prototype optimization

Blade length — proof of concept	1 cm target	Achievable with current technology
Blade length — functional device	~1 meter target	REQUIRES MANUFACTURING DEVELOPMENT
Lattice spacing	Tuned to crystal phonon-photon coupling frequency	[Crystal dependent] REQUIRES CALCULATION

NOTE: Scaling from centimeter to meter blade length is the primary manufacturing challenge. No demonstrated pathway exists at time of writing. This is the single largest unresolved engineering problem in the device.

6.4 Structural Behavior Under Active Condensate

PARAMETER	STATUS
Predicted behavior — active	Polariton condensate pressurizes lattice from within — produces structural rigidity
Predicted behavior — inactive	Lattice returns to unloaded flexible state
Experimental confirmation	NOT DEMONSTRATED at any scale. This is a theoretical prediction requiring experimental validation
Priority	CRITICAL PATH — must be demonstrated at proof-of-concept scale before device development

7. COMPONENT 5 — HILT ASSEMBLY

7.1 Specification

PARAMETER	SPECIFICATION
Housing material	Standard engineering — aluminum, titanium, or composite. No exotic material required.
Crystal mount	Vibration-isolated mount. Prevent phonon feedback from blade impact events disrupting condensate
Coil-crystal alignment	Must maintain precise coil-to-crystal geometry under physical impact. Tolerance [REQUIRES CALCULATION]
Activation interface	Physical switch completing coil drive circuit. No biological interface required.
Grip	Standard. No special material required.

7.2 Power Source

PARAMETER	SPECIFICATION
Primary candidate	High-density solid state battery — continuous discharge regulation
Power requirement baseline	~0.53 W/cm² at crystal face (CW condensation threshold — Wu et al. 2025)
Actual power requirement	[REQUIRES PROTOTYPE MEASUREMENT] — coil efficiency and crystal geometry affect power
Alternative candidates	Capacitive discharge bank; piezoelectric harvesting from grip pressure
Selection method	Prototype measurement required. Power system cannot be fully specified before crystal

8. BLADE INTERACTION MECHANISM

PARAMETER	VALUE / STATUS
Interaction mechanism	Polariton-polariton scattering at contact region between two active condensate blades
Physical basis	Non-zero effective mass of polariton quasiparticle — confirmed property

Polariton effective mass	4-5 orders of magnitude below electron mass — inherent to quasiparticle
Interaction constant	$g = 6E_{\text{ph}} \times a_{\text{ph}}^2 / S$ (Ciuti et al. 1998) — confirmed formula
Force magnitude	Proportional to polariton density at contact — controlled by coil drive power
Demonstration status	Polariton-polariton scattering confirmed in laboratory. Force magnitude at blade scale N/A

9. BLADE LENGTH CONTROL

PARAMETER	VALUE / STATUS
Control mechanism	Coil drive power — higher power increases condensate density and extends coherence
Physical basis	Coherence length scales with condensate density in polariton BEC systems — confirmed
Transfer function	[REQUIRES PROTOTYPE MEASUREMENT] — power-to-length calibration curve must be established
Minimum blade length	[REQUIRES PROTOTYPE MEASUREMENT]
Maximum blade length	Limited by waveguide length and power budget [REQUIRES PROTOTYPE MEASUREMENT]

10. MINIMUM VIABLE PROOF-OF-CONCEPT EXPERIMENT

ITEM	SPECIFICATION
Goal	Demonstrate polariton condensate propagation through SWCNT waveguide at room temperature
Crystal	CsPbBr3 single crystal perovskite microwire
Waveguide	SWCNT composite film on cylindrical substrate — 1 cm length
Coil	Custom solenoid wound to crystal phonon resonance frequency
Drive source	CW source at calibrated frequency
Measurement — condensation	Photoluminescence spectroscopy confirming nonlinear threshold behavior at crystal
Measurement — propagation	Interferometric coherence measurement at waveguide output
Measurement — rigidity	Mechanical probe contact force measurement under active vs. inactive condensate
Success criteria	(1) Condensation confirmed above threshold. (2) Coherence detected at waveguide output
Facility requirement	University condensed matter optics laboratory — standard equipment

11. OPEN ENGINEERING PROBLEMS — PRIORITY ORDER

PRI	PROBLEM	DESCRIPTION	BLOCKING
1	Condensate-waveguide rigidity	Structural rigidity under active condensate has not been demonstrated at any scale	Discoverability
2	Blade waveguide scaling	No manufacturing pathway demonstrated for meter-scale SWCNT blade on lattice	Feasibility
3	Crystal optical quality	Polariton lifetime requires crystal quality exceeding room-temperature coherence length	Observability
4	Emitter nanofabrication	FIB milling of crystal emitter face to required geometry	Mode matching
5	Coil tuning protocol	Standardized procedure for crystal resonance measurement and matched coil winding	Repeatability

12. PRIMARY REFERENCES — VERIFIED AND CORRECTED

All references verified against published literature. One correction applied: the original document contained a phantom citation ('Held et al. 2024 — Nature Photonics') which does not exist. The correct citation for electrical pumping of SWCNT polaritons is Graf et al. 2017 — Nature Materials, confirmed below.

REFERENCE	FULL CITATION	RELEVANCE / STATUS
Su et al. 2018 — Science Advances	Su, C. et al. Room temperature long-range coherent exciton-polariton condensation in CsPbBr ₃ Sci Adv 4, eabg1801 (2018)	Room temperature polariton condensation in CsPbBr ₃
Wu et al. 2025 — Science Advances	Wu, C. et al. Room-temperature continuous-wave pumped exciton-polariton condensation in a perovskite microring resonator Nat Commun 16, 1053 (2025)	Exciton-polariton condensation in a perovskite microring resonator
Kedziorowa et al. 2024 — Nature Materials	Kedziorowa, M. et al. Predesigned perovskite crystal waveguide for the assisted fabrication of ambipolarly shaped polariton devices Nat Mater 23, 1000–1006 (2024)	Predesigned perovskite crystal waveguide for the assisted fabrication of ambipolarly shaped polariton devices
Graf et al. 2016 — Nature Communications	Graf, A. et al. Near-infrared exciton-polaritons in strongly coupled SWCNTs enabled by a holey carbon nanotube Nat Commun 7, 10616 (2016)	Near-infrared exciton-polaritons in strongly coupled SWCNTs enabled by a holey carbon nanotube
Graf et al. 2017 — Nature Materials	Graf, A. et al. CORRECTED: electrical pumping and tuning of Helium-like excitations in SWCNT polaritons Nat Mater 16, 649–654 (2017)	CORRECTED: electrical pumping and tuning of Helium-like excitations in SWCNT polaritons
Ardizzone et al. 2022 — Nature Communications	Ardizzone, V. et al. Polariton Bose–Einstein condensate formation in a microcavity Nat Commun 13, 4452 (2022)	Polariton Bose–Einstein condensate formation in a microcavity
Ciuti et al. 1998 — Physical Review Letters	Ciuti, B., Savona, V. & Quattropani, A. Role of the exchange interaction in the scattering of excitons and polaritons Phys Rev Lett 81, 1757 (1998)	Role of the exchange interaction in the scattering of excitons and polaritons